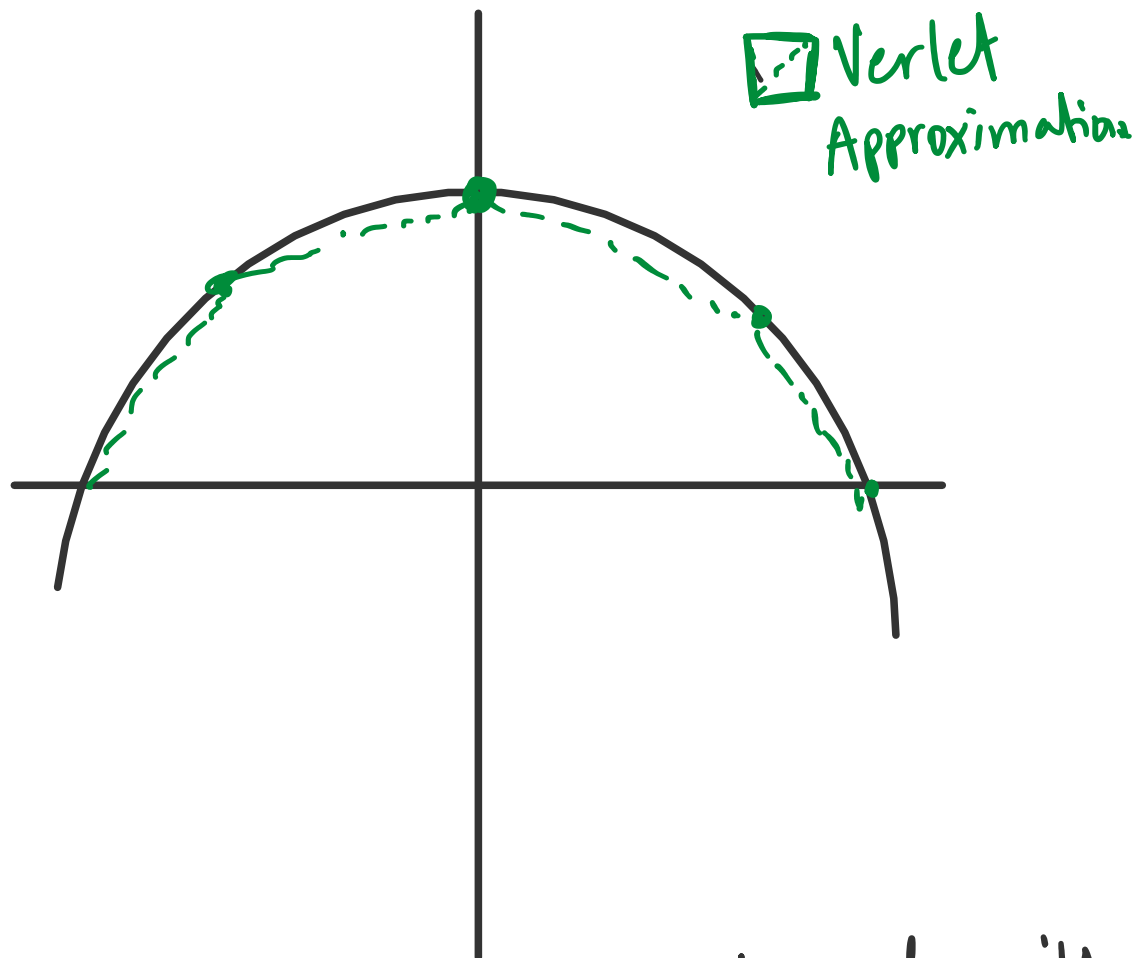


Velocity Verlet (NOTES)

Monday, 20 July 2026

11:40 PM



Instead of tracking velocity and position separately, Verlet uses the current position, and the previous position to figure out the next one. Velocity is a "side effect".

Verlet formula: $x_{\text{next}} = 2x_{\text{current}} + x_{\text{previous}} + ah^2$

Verlet has better energy conservation over long periods of time.

Also much less calculations heavy.

Additionally, there is also a more improved variant: the velocity verlet:

$x_{\text{next}} = x + vh + \frac{1}{2}ah^2$ (Basically one of the kinematics equations)

$v_{\text{next}} = v + \frac{1}{2}(a + a_{\text{next}})h$